WORKED SOLUTION

STEP I 1999, Question 9

Mechanics • Kinematics at Constant Speed

STEP I 1999, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

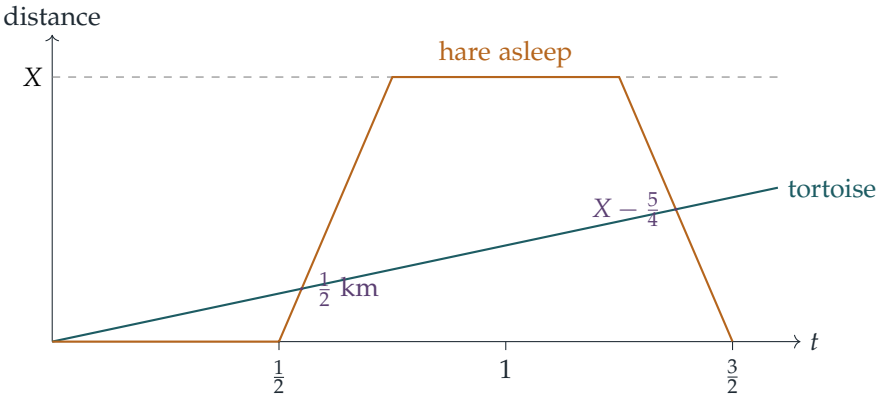
step.maths.org/questions/99-s1-q9

The key idea

Everything happens at constant speeds, so every journey time is just distance over speed. Each time the animals meet they are at the same place at the same moment, so we write down the time on the clock for each of them and set the two equal. Two meetings give two equations in v and V. Working with $1/v$ and $1/V$ (hours per kilometre) keeps the equations linear.

Solution

We model both animals as moving at constant speed along the same straight course, ignoring any time to turn round. We measure time t in hours from the start. A distance–time sketch (drawn here for the value of X found at the end) shows the two crossings we use.



The first meeting

The tortoise reaches the point $\frac{1}{2}$ km from the start at time $\dfrac{1/2}{v}$. The hare starts at $t = \frac{1}{2}$ and reaches the same point at time $\frac{1}{2} + \dfrac{1/2}{V}$. They are there together, so

$$\frac{1}{2v} = \frac{1}{2} + \frac{1}{2V}, \qquad \text{that is} \qquad \frac{1}{v} = 1 + \frac{1}{V}. \tag{1}$$

The second meeting

The hare reaches the vegetable patch at time $\frac{1}{2} + \dfrac{X}{V}$, sleeps for half an hour, and leaves at $1 + \dfrac{X}{V}$. She then runs $\frac{5}{4}$ km back, so she meets the tortoise at time

$$1 + \frac{X}{V} + \frac{5}{4V} = 1 + \frac{X + \frac{5}{4}}{V}.$$

The tortoise is still on the way out, $X - \frac{5}{4}$ km from the start, so she has been walking for $\dfrac{X - \frac{5}{4}}{v}$ hours. Setting the two times equal:

$$\frac{X - \frac{5}{4}}{v} = 1 + \frac{X + \frac{5}{4}}{V}. \tag{2}$$

Ask yourself We have two equations and two unknowns, but (2) has $1/v$ multiplied by X. How do we get V on its own?

We substitute (1) into (2). The X/V terms then cancel, which is what makes V come out so neatly:

$$\begin{aligned}
\left(X - \tfrac{5}{4}\right)\left(1 + \frac{1}{V}\right) &= 1 + \frac{X + \frac{5}{4}}{V} \\
X - \tfrac{5}{4} + \frac{X - \frac{5}{4}}{V} &= 1 + \frac{X + \frac{5}{4}}{V} \\
X - \tfrac{9}{4} &= \frac{5/2}{V}.
\end{aligned}$$

Hence

$$V = \frac{5/2}{X - \frac{9}{4}} = \frac{10}{4X - 9},$$

as required. (For this to be a positive speed we need $X > \frac{9}{4}$, which we keep in mind for the last part.)

Then from (1),

$$\frac{1}{v} = 1 + \frac{4X - 9}{10} = \frac{4X + 1}{10}, \qquad \text{so} \qquad v = \frac{10}{4X + 1}.$$

Finding X

The hare leaves the patch at $1 + \dfrac{X}{V}$ and takes another $\dfrac{X}{V}$ hours to get home, so she finishes at $1 + \dfrac{2X}{V}$. Setting this equal to $\frac{3}{2}$ gives $\dfrac{2X}{V} = \frac{1}{2}$, so $V = 4X$. Therefore

$$\begin{aligned} 4X &= \frac{10}{4X - 9} \\ 16X^2 - 36X - 10 &= 0 \\ 8X^2 - 18X - 5 &= 0 \\ (2X - 5)(4X + 1) &= 0. \end{aligned}$$

The root $X = -\frac{1}{4}$ is impossible for a distance (and anyway we need $X > \frac{9}{4}$), so

$$X = \tfrac{5}{2} \text{ km}.$$

With this value the hare's speed is $V = 10\,\text{km/h}$ and the tortoise's is $v = \frac{10}{11}\,\text{km/h}$.

Checks

Put $X = \frac{5}{2}$, $V = 10$, $v = \frac{10}{11}$ back into the story. The tortoise reaches $\frac{1}{2}$ km at $t = 0.55$, and the hare, starting at $t = 0.5$, needs 0.05 h to get there: they meet. The hare reaches the patch at $t = 0.75$ and leaves at $t = 1.25$. She reaches 1.25 km from the patch (which is 1.25 km from the start) at $t = 1.375$, and the tortoise reaches 1.25 km at $1.25 \times 1.1 = 1.375$: they meet again. The hare gets home at $1.25 + 0.25 = 1.5$. Substituting random values of $X > \frac{9}{4}$ into the formulas for V and v and replaying the race also gives meetings at exactly the right places.

Common slips

- Forgetting one of the hare's half-hour sleeps, especially the one at the patch, when writing her clock time.
- Taking the second meeting point as $\frac{5}{4}$ km from the *start* rather than from the vegetable patch.
- Keeping both roots of the quadratic: a negative distance must be rejected, with a reason.

Beyond STEP

A distance–time graph like the one above turns a word problem into a question about where straight lines cross, which is the starting point for kinematics in any mechanics course. Working with reciprocals of speeds (*pace*) made the equations linear here; choosing variables that linearise a problem is a trick that recurs throughout applied

mathematics. For a first-year treatment of motion that grows from these ideas, try Kleppner and Kolenkow, *An Introduction to Mechanics*.

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